Terminal phases, regularity thresholds, and generic noncapture for AdGD2

Abstract

We study the literal AdGD2 controller with its published initialization. On a scalar quadratic every infinite run is captured in finite time by an admissible nonconstant stepsize two-cycle. For a nonflat scalar minimum, Dini continuity of the Hessian yields limiting controller phases and geometric primal decay; a log-Dini condition yields nonzero pure-geometric amplitudes. We construct a fixed strongly convex C^2 objective with continuous non-Dini Hessian and nonconvergent step parities, and another with a Dini but non-log-Dini Hessian whose geometric amplitudes degenerate through an explicit polynomial prefactor. A source-initialized cubic superlinear orbit realizes the centered controller phase. Finally, on $\mathrm{diag}(1,2)$, for every fixed mixed direction all but countably many initial steps avoid capture and have nonconvergent steps, although the primal iterates converge. These are controller and terminal-rate conclusions, not failures of the source convergence theorem.

Keywords: adaptive gradient descent; terminal dynamics; Dini continuity; superlinear convergence; stepsize nonconvergence

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1 The literal algorithm and scope

For finite convex differentiable f write $g = \nabla f$. AdGD2, Algorithm 2 of Malitsky–Mishchenko [1], starts with

$$x_0, \qquad \alpha_0 > 0, \qquad \theta_0 = 1/3, \qquad x_1 = x_0 - \alpha_0 g(x_0). \tag{1}$$

For $k \geq 1$, before termination, it uses

$$\begin{aligned} L_k &= \frac{\|g(x_k) - g(x_{k-1})\|}{\|x_k - x_{k-1}\|}, \\ \alpha_k &= \min\left\{ \alpha_{k-1}\sqrt{\tfrac{2}{3} + \theta_{k-1}}, \frac{\alpha_{k-1}}{\sqrt{[2\alpha_{k-1}^2 L_k^2 - 1]_+}} \right\}, \\ \theta_k &= \alpha_k/\alpha_{k-1}, \qquad x_{k+1} = x_k - \alpha_k g(x_k). \end{aligned} \tag{2}$$

A positive numerator divided by zero is $+\infty$. Stop at a zero gradient; no secant or controller step is requested afterward. All scalar objectives below have unique minimizer zero. No optional initial calibration and no replacement of the measured secant by a global Lipschitz bound is made.

The source Theorem 2 gives iterate convergence for convex objectives with locally Lipschitz gradient and an attained minimum. We use that theorem where stated. The curvature-branch identity

$$\alpha_{k-1}^{-2} + \alpha_k^{-2} = 2L_k^2 \tag{3}$$

is already printed in its Lemma 6 and is not a new identity here. Earlier scalar terminal-rate analysis showed that all-orders flat minima can support arbitrarily slow last-gradient decay, whereas a finite-order flat minimum $g(x) \sim ax^p$, odd $p \geq 3$, yields eventual growth-branch geometric equivalents. These are background distinctions: the present nonflat results concern $g'(0) > 0$, and the planar result concerns controller nonconvergence. The source's open last-gradient question is not a false theorem, and the examples below do not contradict its iterate or best-objective conclusions.

Put throughout

$$c = \tfrac{2}{3}, \quad q = \frac{1+\sqrt{11/3}}{2}, \quad R^3 - cR - 1 = 0, \quad R > 0,$$

$$\ell = \sqrt{(1+R^{-2})/2}, \quad u = \sqrt{(1+R^2)/2}, \quad C(a) = \frac{a}{\sqrt{2a^2-1}} \quad (a > 1/\sqrt{2}). \tag{4}$$

The cubic has exactly one positive root and $R > 1$.

2 Scalar quadratic: global finite capture

Theorem 1. *For $f(x) = x^2/2$, $x_0 \neq 0$ and arbitrary $\alpha_0 > 0$, every infinite run of (1)–(2) is eventually an exact curvature two-cycle. Its unordered step pair is*

$$\{A, B\}, \quad 1 < A \leq u, \quad B = C(A) \in [\ell, 1), \quad A^{-2} + B^{-2} = 2.$$

If $j \geq 2$ is the first curvature index among indices at least two, then it exists and a capture index $m \leq j+2$ satisfies $\alpha_{k+2} = \alpha_k$ for $k \geq m$. Ties count as curvature. Every such unordered pair is attained by $\alpha_0 = A, \theta_0 = 1/3$. After capture,

$$x_{k+2} = (1-A)(1-B)x_k, \qquad -1 < (1-A)(1-B) < 0.$$

The alternative is finite termination, including any occurrence of $\alpha_k = 1$.

Proof. For a nonstopping run the secants equal one and successive points are distinct. Since $\theta_k \leq \sqrt{c+\theta_{k-1}}$ and $q^2 = c+q$, source initialization gives $0 < \theta_k < q$. For fixed $t = c+\theta$, the increasing growth proposal $a\sqrt{t}$ and decreasing finite curvature proposal $C(a)$ meet at $a^2 = (1+1/t)/2$. Thus

$$\alpha_k < U := \sqrt{(1+c+q)/2} < 5/4 \quad (k \geq 1).$$

The last inequality follows from $q < 35/24$, verified by its quadratic equation. If every step from index two onward were growth, its ratios would increase to $q > 1$, making steps unbounded. Thus the stated j exists and its input belongs to $(1/\sqrt{2}, 5/4)$.

The map C is a decreasing involution, with $a^{-2} + C(a)^{-2} = 2$. For a proposed pair $a, b = C(a)$, put $s = b/a$. Both reversals are allowed exactly when

$$1/s \le \sqrt{c+s}, \qquad s \le \sqrt{c+1/s}.$$

These are equivalent to $1/R \le s \le R$, hence to $\ell \le a \le u$. Accordingly any curvature step with input in $[\ell, u]$ is captured.

For an input $h \in (u, 5/4)$ its output $b = C(h)$ lies in $(5/\sqrt{34}, \ell)$. The next selected step is the growth correction

$$d = D(b) := b\sqrt{c + \sqrt{2b^2 - 1}}.$$

Indeed D is increasing, $D(\ell) = u$, and $d < u < h = C(b)$. Also

$$d^2 > 50/51 > \ell^2,$$

since $\sqrt{2b^2-1} > 4/\sqrt{34} > 2/3$, and $R > 6/5$ gives $\ell^2 < 61/72$. Here $d > b$, so the next growth proposal is greater than $\sqrt{250/153}$ while $C(d) < \sqrt{50/49}$. The next step is therefore curvature from an admissible input d, giving capture at $m = j + 1$.

If instead the first curvature input $a < \ell$, its output $v = C(a) > u$ still satisfies $v < U < 5/4$. Its incoming ratio is $v/a > 1$, so its next growth proposal exceeds v, while its curvature proposal is $a < v$. It returns to a by curvature. Apply the preceding correction to this oversized input v; capture occurs at $m = j+2$. This exhausts the cases. A captured value one terminates the actual optimization run and so cannot occur in the infinite case.

Every nonconstant pair is reached by starting at its larger value A: the first growth proposal is A, the curvature proposal is $B < A$, and the admissibility inequalities then apply forever. The primal pair multiplier is negative because the two steps straddle one, and greater than -1 since both lie in $(0, 5/4)$. This proves all assertions. ∎

3 Nonflat minima and Dini phase limits

For a local Hessian modulus use a nondecreasing $\omega(t) \to 0$ with $|g'(x) - a| \le \omega(|x|)$ near zero, where $a = g'(0) > 0$. Call it Dini if

$$\int_0^\delta \omega(t)\,\frac{dt}{t} < \infty,$$

and log-Dini if the same integral with the additional factor $1 + |\log t|$ is finite.

Theorem 2. *Let $f : \mathbb{R} \to \mathbb{R}$ be finite, convex and C^1, with locally Lipschitz derivative g, unique minimizer zero, and $g \in C^1$ near zero with $a = g'(0) > 0$. Suppose its local Hessian modulus is Dini. Every source-initialized run either terminates or satisfies*

$$|x_k| \le C_0 \rho^k \quad (0 < \rho < 1), \qquad \alpha_{2n} \to A, \quad \alpha_{2n+1} \to B,$$

$$aA, aB \in [\ell, u], \qquad A^{-2} + B^{-2} = 2a^2, \qquad x_{k+2}/x_k \to \mu = (1 - aA)(1 - aB).$$

If $A \neq B$, then $-1 < \mu < 0$. If $A = B$, both equal $1/a$ and $|x_{k+1}/x_k| \to 0$. Under a locally Hölder Hessian of any positive order, step parity convergence is exponential. Under a log-Dini Hessian and $A \neq B$,

$$x_{2n}/\mu^n \to C_0' \neq 0, \qquad x_{2n+1}/\mu^n \to C_1' = (1 - aA)C_0' \neq 0. \tag{5}$$

A locally Lipschitz Hessian gives both of these stronger conclusions.

Proof. The source convergence theorem gives $x_k \to 0$, hence $L_k \to a$. The universal ratio bound is $\theta_k \leq q$. Eventually $L_k \geq a(1 - \eta)$, and the crossing argument from Theorem 1 gives $\alpha_k \leq U/[a(1 - \eta)]$, with $U < 5/4$.

There is also an eventual positive lower bound. Let $L_{\max}$ bound local secants. Every curvature output exceeds $1/(\sqrt{2}L_{\max})$. A growth step can decrease only when its incoming ratio is below $1/3$, which must follow curvature because a growth-selected ratio is at least $\sqrt{c}$. Such a decreasing step exceeds $1/(\sqrt{3}L_{\max})$. Including finitely many initial local steps gives $\alpha_k \geq b > 0$. Since $g(x)/x \to a$, eventually

$$0 < \delta \leq \alpha_k g(x_k)/x_k \leq H < 2.$$

Thus $|x_{k+1}| \leq \rho|x_k|$ for $\rho = \max\{|1 - \delta|, |1 - H|\} < 1$. The integral formula for the secant now gives

$$\varepsilon_k := |L_k/a - 1| \leq a^{-1}\omega(C\rho^{k-1}), \qquad \sum_k \varepsilon_k < \infty.$$

The summability follows by integration on geometric annuli. Hölder continuity makes these errors exponentially small.

To handle ties without assuming eventual curvature selection, use the following total map of the constant curvature controller

$$T(h, r) = (F(h, r), F(h, r)/h), \quad F(h, r) = \left[\max\left\{\frac{1}{h^2(c + r)}, 2 - \frac{1}{h^2}\right\}\right]^{-1/2}$$

on $h > 0, 0 < r \leq q$. This is a locally Lipschitz map, including at branch ties. Unlike the stopping optimization algorithm, this auxiliary map continues at $h = 1$. Its cycle set is

$$S = \{(h, h/C(h)) : \ell \leq h \leq u\}, \qquad T^2|_S = I.$$

In particular $(1, 1) \in S$, but $(1, r)$ with arbitrary r need not be fixed.

On every compact rectangle D in this domain, a uniformly bounded number of iterations enters S. Here are the uniform details. After one step $h \leq U$. If steps from index two onward remain growth, two growth iterations make their ratios at least $\sqrt{c + \sqrt{c}} > 1$. The first output has a positive lower bound on D, so sufficiently many such growth steps contradict $h \leq U$. A curvature step with index at least two therefore occurs uniformly

soon. The admissible-input, oversized-input and undersized-input arguments of Theorem 1 place its output in S after at most three additional updates. Those arguments remain valid for the total map even if a value is one. Thus for an even E sufficiently large, $P = T^E$ maps D to S, is locally Lipschitz, and obeys

$$P(T^2 z) = P(z), \qquad P|_S = I.$$

The invariance follows from $T^{E+2}z = T^2(Pz) = Pz$, so it does not require $T^2 z \in D$.

The actual states $z_k = (a\alpha_k, \theta_k)$ eventually lie in a compact positive rectangle. Their map replaces the second entry in the maximum by $2(L_{k+1}/a)^2 - 1/h^2$. Compact-set Lipschitz estimates therefore give

$$\|z_{k+1} - Tz_k\| \le C\varepsilon_{k+1}, \quad \|Pz_{k+2} - Pz_k\| \le C(\varepsilon_{k+1} + \varepsilon_{k+2}).$$

Both parities of Pz_k converge. Comparing E actual and ideal steps gives

$$\|z_{k+E} - Pz_k\| \le C \sum_{i=1}^{E} \varepsilon_{k+i} \to 0.$$

Thus $\operatorname{dist}(z_k, S) \to 0$, and $\|z_k - Pz_k\| \to 0$. The actual parities converge in S and their limits are interchanged by T. This proves the phase interval and inverse-square relation, without a finite-switching claim for variable curvature.

The exact primal pair ratio is

$$\frac{x_{k+2}}{x_k} = \left(1 - \alpha_k \frac{g(x_k)}{x_k}\right)\left(1 - \alpha_{k+1} \frac{g(x_{k+1})}{x_{k+1}}\right).$$

Its limit is μ. Distinct paired phases straddle $1/a$, so $-1 < \mu < 0$; coincident phases equal $1/a$ and imply superlinearity.

For the refined estimates, the preceding projection argument gives, with $z^{(k \bmod 2)}$ the actual parity limit,

$$\|z_k - z^{(k \bmod 2)}\| \le C \sum_{j \ge k-E+1} \varepsilon_j. \tag{6}$$

This proves exponential phase convergence for exponentially small errors. Under log-Dini continuity, geometric-annulus comparison yields $\sum (k+1)\varepsilon_k < \infty$. Summing (6) then proves summable phase errors. Also $|g(x_k)/x_k - a| \le \omega(C\rho^k)$ is summable. Consequently the pair multipliers $M_n = x_{2n+2}/x_{2n}$ satisfy $\sum |M_n - \mu| < \infty$. For $\mu \ne 0$, no factor vanishes on a nonstopping run, and the eventually positive product $\prod (M_n/\mu)$ has a finite nonzero limit. Including finitely many initial signed factors proves (5). Dini continuity alone does not make the tails in (6) summable, which is why that assumption alone is not used for amplitudes. ∎

4 A realized centered phase with exact superlinear order

Theorem 3. *Under the locally Lipschitz-Hessian hypotheses of Theorem 2, suppose*

$$g(x) = ax + bx^p + o(|x|^p), \quad p \geq 2 \text{ an integer}, \quad b \neq 0.$$

For a nonstopping run with $\alpha_k \to 1/a$, curvature is eventually selected strictly and

$$\alpha_k = \frac{1}{a} - \frac{2b}{a^2} x_k^{p-1} + o(|x_k|^{p-1}), \qquad x_{k+1}/x_k^p \to b/a.$$

This centered phase is realized by a source-initialized nonstopping run. Specifically, define, with $t = |x|$,

$$f(x) = \begin{cases} x^2/2 - x^4/4, & t \leq 1/10, \\ 199/40000 + (99/1000)(t - 1/10) + (97/200)(t - 1/10)^2, & t \geq 1/10. \end{cases}$$

For every $0 < h \leq 1/100$, some

$$\alpha_0 \in [(1 - 3h^2)^{-1/2}, (1 - 5h^2)^{-1/2}]$$

with $x_0 = h, \theta_0 = 1/3$ yields strict curvature selection for all $k \geq 1$, no termination, $\alpha_k \to 1$, and $x_{k+1}/x_k^3 \to -1$.

Proof. In the centered case $t_k = x_{k+1}/x_k \to 0$. The polynomial secant identity and the remainder bound, without differentiating that remainder, give

$$L_{k+1} = a + bx_k^{p-1} + o(|x_k|^{p-1}).$$

Indeed $|x_{k+1} - x_k| \geq |x_k|/2$ eventually, so the remainder difference divided by the secant denominator is $o(|x_k|^{p-1})$. The growth and curvature ratios tend respectively to $\sqrt{5/3}$ and one, proving eventual strict curvature. Set $e_k = \alpha_k^{-2} - a^2$ and $q_j = L_j^2 - a^2$. Then $e_k + e_{k-1} = 2q_k$ and

$$q_{k+1} = 2abx_k^{p-1} + o(|x_k|^{p-1}).$$

Superlinearity implies $\sum_{j \geq k+2} |q_j| = o(|x_k|^{p-1})$. Iterating the recurrence into the future using $e_k \to 0$ gives

$$e_k = 2 \sum_{j \geq k+1} (-1)^{j-k-1} q_j = 4abx_k^{p-1} + o(|x_k|^{p-1}).$$

Taylor expansion of $(a^2 + e_k)^{-1/2}$ and the primal update prove the claimed order and coefficient.

For the explicit objective, the value, derivative and second derivative match at $1/10$: they are $199/40000, 99/1000, 97/100$. Globally $97/100 \leq g' \leq 1$, and g' is Lipschitz. Fix h and let z_0 range over $I_h = [1 - 5h^2, 1 - 3h^2]$. Define an auxiliary polynomial continuation

$$x_{k+1} = x_k[1 - (1 - x_k^2)/\sqrt{z_k}], \quad x_0 = h,$$

$$Q_{k+1} = x_k^2 + x_k x_{k+1} + x_{k+1}^2, \quad \widetilde{L}_{k+1} = 1 - Q_{k+1},$$
$$z_{k+1} = 2\widetilde{L}_{k+1}^2 - z_k.$$

This continuation is used only to prove existence of an initial step; it is not an instruction to continue the stopping algorithm at zero. Put $q_h = 20h^2 \le 1/500$. Induction gives

$$|x_k| \le h q_h^k, \qquad |z_k - 1| \le 20h^2.$$

On $[0.998, 1.002]$, $|(z^{-1/2})'| < 0.6$ and $z^{-1/2} < 1.1$, so the primal factor is at most $13.1h^2 < q_h$. Also $0 \le Q_j \le 3h^2 q_h^{2j-2}$ and $|\widetilde{L}_j^2 - 1| \le 2Q_j$, whence

$$|z_k - 1| \le 5h^2 + 2\sum_{j=1}^{k} |\widetilde{L}_j^2 - 1| \le [5 + 12/(1 - q_h^2)]h^2 < 18h^2.$$

This closes the induction, ensures positivity, and gives finite-step continuity in z_0.

Set $\alpha_k = z_k^{-1/2}$. At the first step, $\widetilde{L}_1^2 \ge 1 - 2h^2(1 + q_h + q_h^2) > 1 - 3h^2 \ge z_0$, so $z_1 > z_0$ and curvature is strictly preferred to the source growth proposal α_0. Later, all step ratios are between 0.98 and 1.02; the growth ratio exceeds $\sqrt{2/3 + 0.98} > 1.28$ and the curvature ratio is below 1.02. Hence curvature is always strictly chosen until any hypothetical zero hit.

With $E_j = \widetilde{L}_j^2 - 1$, the uniformly convergent series

$$P(z_0) = \lim_n z_{2n} = z_0 + 2\sum_{j \ge 1} (-1)^j E_j$$

is continuous. Its odd phase is $2 - P$. The explicit bound

$$|P(z_0) - z_0 - 4h^2| \le 100h^4$$

follows from $|Q_1 - h^2| \le h^2(q_h + q_h^2)$: the first summand error is below $83h^4$, and the remaining sum is below $0.481h^4$. Thus $P(1 - 5h^2) < 1 < P(1 - 3h^2)$. The intermediate value theorem gives $z_0^* \in I_h$ with $P(z_0^*) = 1$.

This root cannot terminate. If N is the first zero hit and $y = x_{N-1} \ne 0$, the preceding update forces $z_{N-1} = (1 - y^2)^2 < 1$ and $\widetilde{L}_N = 1 - y^2$, hence $z_N = z_{N-1} < 1$. Later auxiliary points are zero and later z values alternate exactly between z_N and $2 - z_N$, contradicting both phase limits being one. The root therefore defines a genuine infinite source-initialized run. Applying the first part with $a = 1, b = -1, p = 3$ finishes the proof. No uniqueness or closed form for this initial step is asserted. ∎

5 A common fixed-function interpolation construction

We next construct two objectives, using one template that verifies all primal nodes, secants and controller choices. Let a sequence (v_n) lie in $[0.84, 0.88]$ with $v_n - v_{n-1} \to 0$,

and set

$$z_{2n} = v_n, \quad z_{2n+1} = 2 - v_n, \quad A_k = z_k^{-1/2}, \quad \lambda_k = \sqrt{(z_k + z_{k-1})/2} \quad (k \geq 1).$$

For the two sequences below, $0.989 < \lambda_k < 1.011$ and $\lambda_k \to 1$, while $\lambda_{2n+1} = 1$. Define

$$G_0 = 1, \quad G_{k+1} = (1 - A_k\lambda_{k+1})G_k, \quad X_k = \sum_{j=k}^{\infty} A_j G_j, \quad E_k = G_k - X_k.$$

At even k, $-0.1 < 1 - A_k\lambda_{k+1} < -0.06$; at odd k, $0.03 < 1 - A_k\lambda_{k+1} < 0.1$. Thus

$$0.03 < |G_{k+1}/G_k| < 0.1, \tag{7}$$

and the signs repeat $+, -, -, +, +, -, -, +, \ldots$. Tails converge absolutely and yield exact identities

$$\begin{aligned} X_{k+1} &= X_k - A_k G_k, \\ G_k &= \sum_{j=k}^{\infty} A_j \lambda_{j+1} G_j, \\ E_k &= \sum_{j=k}^{\infty} A_j (\lambda_{j+1} - 1) G_j. \end{aligned} \tag{8}$$

Since $A_k < 1.1$,

$$|E_k/G_k| \leq \tfrac{11}{9} \sup_{j \geq k} |\lambda_{j+1} - 1| < 0.014.$$

Consequently

$$0.986 < X_k/G_k < 1.014, \quad |E_k/X_k| < 0.015, \quad 1/40 < |X_{k+1}/X_k| < 1/9. \tag{9}$$

Every node is nonzero, all are distinct, and their only finite accumulation point is zero. Consecutive same-sign nodes differ by at most three temporal indices, so their magnitudes $r > s$ obey $r/40^3 < s < r/9$. Most importantly,

$$\frac{G_k - G_{k-1}}{X_k - X_{k-1}} = \lambda_k > 0. \tag{10}$$

Define one residual function e on the entire real line by $e(0) = 0$, $e(X_k) = E_k$ and, between adjacent same-sign nodes $b < d$, by

$$e(x) = E_b + (E_d - E_b) P\left(\frac{x - b}{d - b}\right), \quad P(t) = 3t^2 - 2t^3.$$

Here E_b, E_d denote their assigned residual values. Extend constantly outside the two outermost nodes. Since $P'(0) = P'(1) = 0$, the derivatives join. On every interpolation interval,

$$|e'(x)| \leq \tfrac{3}{2}(0.015)\frac{r + s}{r - s} \leq 0.028125.$$

Moreover $E_k/X_k \to 0$. The bounded same-side mesh ratio gives $e(x)/x \to 0$ at zero; the derivative bound with the endpoint relative errors replacing 0.015 gives $e'(x) \to 0$. Hence $e \in C^1(\mathbb{R})$ and $e'(0) = 0$. Put

$$g(x) = x + e(x), \qquad f(x) = \int_0^x g(t)\,dt. \tag{11}$$

This fixed function is globally C^2, strongly convex, has globally Lipschitz gradient, and satisfies

$$0.971875 \le f''(x) \le 1.028125, \quad f''(0) = 1, \quad g(X_k) = G_k.$$

Initialize $x_0 = X_0, \alpha_0 = A_0, \theta_0 = 1/3$. The identities (8)–(10) make $L_k = \lambda_k$, and

$$2A_{k-1}^2\lambda_k^2 - 1 = z_k/z_{k-1} > 0,$$

so the curvature proposal is exactly A_k. At $k = 1$, $A_1 < A_0$ and growth equals A_0. At later indices strict curvature selection is equivalent to

$$z_k > \frac{z_{k-1}}{2/3 + \sqrt{z_{k-2}/z_{k-1}}}.$$

For even k the right side is at most $1.16/(2/3+\sqrt{0.84/1.16}) < 0.77 < 0.84$; for odd $k \ge 3$ it is at most $0.88/(2/3+\sqrt{1.12/0.88}) < 0.50 < 1.12$. Thus induction verifies the literal algorithm forever on this one objective, with $x_k = X_k, \alpha_k = A_k$. No externally prescribed secants or post-termination steps are involved.

6 Continuous non-Dini Hessian: neither step parity converges

Theorem 4. *There is a globally strongly convex C^2 function with globally Lipschitz gradient and $f''(0) = 1$ whose source-initialized nonstopping AdGD2 trajectory converges geometrically to zero, while neither stepsize parity converges. Its Hessian is continuous but not Dini at zero, and every controller choice is strictly curvature-selected.*

Proof. In Section 5 take

$$v_n = 43/50 + (1/50)\sin\log(n+1).$$

It lies in $[0.84, 0.88]$, its differences tend to zero, and $\lambda_{2n} = \sqrt{1 + (v_n - v_{n-1})/2} \in (0.989, 1.011)$. Subsequences of integers asymptotic to $\exp(\pi/2+2\pi j)$ and $\exp(3\pi/2+2\pi j)$ show that v_n has distinct limits $0.88, 0.84$. Therefore neither $\alpha_{2n} = v_n^{-1/2}$ nor $\alpha_{2n+1} = (2 - v_n)^{-1/2}$ converges. The template proves every objective, initialization, branch and primal statement.

For non-Dini regularity, let $\omega(r) = \sup_{|x|\le r} |g'(x) - 1|$. The secant integral and (9) imply

$$|\lambda_k - 1| \le \omega(|X_{k-1}|) \le \omega(|X_0| 9^{-(k-1)}).$$

A finite Dini integral would make the sampled right side summable. But

$$|\lambda_{2n} - 1| = \frac{|v_n - v_{n-1}|}{2(\lambda_{2n} + 1)}.$$

The total variation of (v_n) is infinite, since finite total variation would imply convergence. The denominator is uniformly bounded, a contradiction. This proves the failure of Dini continuity for the actual interpolated Hessian. ∎

Continuity alone therefore cannot replace the Dini hypothesis in the phase theorem. This does not claim failure for every non-Dini modulus or a necessary-and-sufficient regularity classification.

7 Dini but non-log-Dini: vanishing geometric amplitudes

Theorem 5. *There is a globally strongly convex C^2 function with globally Lipschitz gradient and Dini Hessian, but non-log-Dini Hessian, whose source-initialized run has distinct limiting phases A, B and*

$$x_{2n} \sim K\mu_*^n n^\kappa, \qquad x_{2n+1} \sim (1 - A)K\mu_*^n n^\kappa,$$

where $K > 0$, $-1 < \mu_* < 0$ and $\kappa < 0$. Thus both pure-geometric quotients tend to zero, rather than to nonzero amplitudes.

Proof. Use the template with $v_* = 43/50$, $c_* = 1/100$ and

$$v_n = v_* + \frac{c_*}{n+1}, \qquad \lambda_{2n} = \sqrt{1 - \frac{c_*}{2n(n+1)}} \quad (n \ge 1).$$

The template again proves all global function and literal-controller facts. Rationalization gives

$$|\lambda_{2m} - 1| \le c_*/(m+1)^2, \qquad \sup_{j \ge k} |\lambda_{j+1} - 1| \le 4c_*/(k+1)^2.$$

Thus (8)–(9) give $|E_k/X_k| < 1/[20(k+1)^2]$. On an interpolation interval whose outer endpoint has temporal index i,

$$|e'(x)| \le \frac{3}{32(i+1)^2}.$$

Every interval meeting $|x| \le |X_k|$ has outer index at least $k - 3$, so for $k \ge 3$ the true radial modulus satisfies

$$\omega(|X_k|) \le \frac{3}{32(k-2)^2}.$$

Each annulus $[|X_{k+1}|, |X_k|]$ has logarithmic length below $\log 40$. Summing its integral bound proves the Dini condition on the entire interpolated Hessian, not just its sampled secants. In particular $\omega(r) = O((1 + \log(|X_0|/r))^{-2})$ for small r.

Define

$$a(v) = 1 - v^{-1/2}, \quad b(v) = 1 - (2 - v)^{-1/2}, \quad \mu(v) = a(v)b(v),$$

$$A = v_*^{-1/2}, \quad B = (2 - v_*)^{-1/2}, \quad \mu_* = \mu(v_*), \quad \kappa = \frac{c_* \mu'(v_*)}{\mu_*}.$$

The exact pair gradient factor is

$$R_n = \frac{G_{2n+2}}{G_{2n}} = a(v_n) \left[1 - (2 - v_n)^{-1/2} \sqrt{1 + (v_{n+1} - v_n)/2}\right].$$

Smoothness on the fixed phase interval yields

$$R_n = \mu_* + \frac{c_* \mu'(v_*)}{n + 1} + O((n + 1)^{-2}).$$

Every $R_n < 0$, so $q_n = R_n/\mu_* > 0$ and

$$\log q_n = \frac{\kappa}{n + 1} + d_n, \qquad \sum |d_n| < \infty.$$

If $\gamma = \lim_n (\sum_{j=1}^{n} 1/j - \log n)$, the exact product gives

$$K = \exp\left(\kappa\gamma + \sum_{j \geq 0} d_j\right) > 0, \qquad G_{2n}/(\mu_*^n n^\kappa) \to K.$$

Since $X_k/G_k = 1 + O((k + 1)^{-2})$, the even primal equivalent follows, and the odd one follows from the one-step factor. Finally

$$\frac{\mu'(v)}{\mu(v)} = -\frac{1}{2v(1 - \sqrt{v})} - \frac{1}{2(2 - v)(\sqrt{2 - v} - 1)} < 0 \quad (0 < v < 1),$$

so $\kappa < 0$.

For non-log-Dini regularity put $r_k = |X_k|$. The actual secant implies

$$\omega(r_{2n-1}) \geq |\lambda_{2n} - 1| = \frac{c_*}{2n(n + 1)(\lambda_{2n} + 1)} \geq \frac{c_*}{4n(n + 1)}.$$

On each disjoint interval $[r_{2n-1}, r_{2n-2}]$, the logarithmic length exceeds $\log 9$ and $1 + |\log r|$ is bounded below by a fixed positive multiple of n for large n. Its weighted integral contribution is therefore at least a positive constant times $1/n$. Summation proves divergence of the log-Dini integral. ∎

This example separates convergence of step parities and pair ratios from nonzero asymptotic amplitudes. It complements, rather than contradicts, Theorem 2.

8 A quantitative nonlinear phase-selection law

For an explicit local illustration, take $f(x) = x^2/2 + x^4/4$ and $\alpha_0 = a = 21/20$. Define

$$s_* = 400/441, \quad b = C(a), \quad p = 1 - a, \quad q_0 = 1 - b, \quad H(t) = 1 + t + t^2.$$

Proposition 8.1. For $0 < |x_0| \le 1/100$ and $\theta_0 = 1/3$, the run never terminates and always chooses curvature strictly. Its even inverse-square phase satisfies

$$S(x_0) := \lim_n \alpha_{2n}^{-2} = s_* + \mathcal{C}x_0^2 + \mathcal{R}(x_0),$$

$$\mathcal{C} = \frac{4[-H(p) + p^2 H(q_0)]}{1 - (pq_0)^2} < -3.78, \qquad |\mathcal{R}(x_0)| \le 4900|x_0|^4.$$

In particular $S(x_0) < s_*$ on this punctured interval. It is even and extends continuously at zero with value s_*, without continuing the stopping algorithm at zero.

Proof. Write $t = x_0$, $s_k = \alpha_k^{-2}$ and $\bar{s}_k = s_*$ for even k, $2 - s_*$ for odd k. The exact secant is $L_k = 1 + Q(x_k, x_{k-1})$, where $Q(u, v) = u^2 + uv + v^2$. On curvature,

$$s_k = 2 - s_{k-1} + \delta_k, \qquad \delta_k = 4Q(x_k, x_{k-1}) + 2Q(x_k, x_{k-1})^2.$$

Joint induction gives

$$|x_k| \le |t|0.1^k, \quad |s_k - \bar{s}_k| \le 12.124t^2, \quad 0.9 \le s_k \le 1.1.$$

Indeed the proposed increment satisfies $0 \le \delta_j \le 12.0018t^2\, 0.01^{j-1}$, whose sum is below $12.124t^2$. Both phase neighborhoods remain within $[0.9, 1.1]$, so proposed steps are between 0.95 and 1.06 and all curvature radicands are positive. The first curvature proposal is below one, less than the growth proposal a. Subsequently growth exceeds $0.95\sqrt{2/3 + 0.95/1.06} > 1.18 > 1.06$. Every accepted step contracts the primal magnitude by less than 0.1. The initial factor is $-0.05 - 1.05t^2$. Even steps exceed one, and odd steps are below 0.96, so no factor vanishes for $|x| \le 0.01$. This proves the induction for the actual algorithm.

Absolute summability gives the exact series $S(t) = s_* + \sum_{j\ge1}(-1)^j\delta_j$. Let $u_0 = 1$, $u_{k+1} = (1 - \bar{s}_k^{-1/2})u_k$; then $u_{2n} = (pq_0)^n$, $u_{2n+1} = p(pq_0)^n$. For $y_k = tu_k$, the derivative bound $|(s^{-1/2})'| < 0.6$ on $[0.9, 1.1]$ gives $|\alpha_k - \bar{s}_k^{-1/2}| < 8t^2$. Subtraction of the exact recurrences yields

$$|x_{k+1} - y_{k+1}| \le 0.05|x_k - y_k| + 10|t|^3 \cdot 0.1^k,$$

so $|x_k - y_k| \le 200|t|^3 \cdot 0.1^k$ uniformly in k. Factoring differences of the three terms of Q gives

$$|Q(x_j, x_{j-1}) - Q(y_j, y_{j-1})| \le 1200|t|^4 \cdot 0.01^{j-1}.$$

Therefore

$$\delta_j = 4t^2 Q(u_j, u_{j-1}) + r_j(t), \quad |r_j(t)| \le 4800|t|^4 \cdot 0.01^{j-1} + 18|t|^4 \cdot 0.0001^{j-1}.$$

The total remainder is below $4900|t|^4$, justifying passage from finite partial sums to the infinite phase expansion. The leading series groups into

$$4\sum_{j\geq 1}(-1)^j Q(u_j, u_{j-1}) = 4[-H(p) + p^2 H(q_0)] \sum_{n\geq 0}(pq_0)^{2n} = \mathcal{C}.$$

Since $H(p) = 381/400$, $p^2 = 1/400$ and $0 < q_0 < 0.05$ gives $H(q_0) < 3$, one has $\mathcal{C} < -3.78$. Thus $S(t) - s_* \leq (-3.78 + 4900t^2)t^2 < 0$. The odd gradient and even secant imply $x_k(-t) = -x_k(t)$ and equal steps, proving evenness. The uniform remainder proves continuity of the extension. No strict monotonicity in $|t|$ is asserted. ∎

9 Generic noncapture on a planar quadratic

Consider $f(x) = (x_1^2 + 2x_2^2)/2$. For a mixed point define

$$r_n = \left(\frac{2x_{n,2}}{x_{n,1}}\right)^2, \quad a_n = \alpha_n, \quad l(r) = \frac{1+4r}{1+r}, \quad q_1(a) = \left(\frac{1-2a}{1-a}\right)^2.$$

The literal old-gradient secant has $L_{n+1}^2 = l(r_n)$, because the displacement is proportional to $(x_{n,1}, 2x_{n,2})$. The projected recurrence before extinction is

$$\begin{aligned} r_{n+1} &= r_n q_1(a_n), \\ G_n &= a_n\sqrt{2/3 + \theta_n}, \quad C_n = [2l(r_n) - a_n^{-2}]_+^{-1/2}, \\ a_{n+1} &= \min\{G_n, C_n\}, \qquad \theta_{n+1} = a_{n+1}/a_n. \end{aligned} \tag{12}$$

A nonpositive curvature radicand means $C_n = +\infty$. A coordinate is extinguished exactly at $a_n = 1$ or $a_n = 1/2$.

Theorem 6. *For every fixed $r_0 = \rho > 0$, there is an at most countable set $B_\rho \subset (0, \infty)$ such that for every $\alpha_0 \notin B_\rho$, every mixed x_0 realizing this direction, and $\theta_0 = 1/3$, neither coordinate is ever extinguished, no finite step equals $2/3$, and (α_n) does not converge. Nevertheless $x_n \to 0$. Also these conclusions hold for planar-almost every (r_0, α_0) and three-dimensional-almost every (x_0, α_0).*

9.1 Finite-word noncollapse on the actual initialization surface

We give a complete algebraic argument, which also proves the fixed-direction strengthening. A finite word in growth and curvature determines an algebraic recurrence. Work first over $\mathbb{C}(u, t)$, with independent variables $u = r_0, t = a_0$, and $\theta_0 = 1/3$. Implement the word by

$$r' = r\frac{(1-2a)^2}{(1-a)^2}, \qquad \left(\frac{a'}{a}\right)^2 = \begin{cases} 2/3 + \theta, & \text{growth}, \\ (2l(r)a^2 - 1)^{-1}, & \text{curvature}, \end{cases} \qquad \theta' = a'/a.$$

Allow either algebraic square-root sign. Any analytic germ of a physically realized finite word belongs to one such finite algebraic field extension.

Extend the t-adic valuation to this finite extension, normalized by $v(t) = 1$. Such an extension exists for valuations on finite algebraic field extensions. We claim inductively that $v(a_j) = 1$, r_j has residue u, and θ_j has finite nonzero residue z_j. Initially this is true. The r multiplier has residue one. On curvature the denominator $2l(r_j)a_j^2 - 1$ has residue -1, so the new ratio has residue $+i$ or $-i$. On growth the only obstruction would be $z_j = -2/3$.

That obstruction never occurs, for any algebraic signs. Put $p(z) = z^2 - 2/3$. A chain of growth residues starting at $1/3$ or $\pm i$ could reach $-2/3$ only if $p^m(-2/3)$ equaled its starting residue for some $m \geq 0$. These iterates are real rational, excluding $\pm i$. For $m \geq 1$ their reduced denominators are exactly 3^{2^m}: $p(-2/3) = -2/9$, and if $b/3^d$ is reduced then

$$p(b/3^d) = \frac{b^2 - 2 \cdot 3^{2d-1}}{3^{2d}},$$

whose numerator is not divisible by three. Thus they cannot equal $1/3$; $m = 0$ also does not. This proves the induction. Every finite-word step has valuation one, so it cannot identically equal any nonzero constant.

The same proof works over $\mathbb{C}(t)$ for every fixed $u = \rho > 0$. The residue of r_j is now ρ, and $1 + \rho \neq 0$ ensures $l(r_j)$ has finite residue. No real continuation of a selected radical branch to $t = 0$ is asserted. The valuation is algebraic, so a physical branch may switch or cease to exist along that real path without affecting the argument.

For each finite word w and $c_0 \in \{1/2, 1, 2/3\}$, let S_{w,c_0} consist of initial data realizing that word through a_n, without earlier extinction, and satisfying $a_n = c_0$. These are semialgebraic sets: introduce finitely many positive intermediate variables, encode squared recurrences and branch inequalities by polynomial equalities and inequalities, clear nonzero denominators, and project. Ties may be assigned either letter. Every selected curvature radical is positive.

If S_{w,c_0} in the two-dimensional source plane had positive measure, a semialgebraic set of that dimension would contain an open ball. Positive-root formulas there give analytic germs and a finite algebraic extension of $\mathbb{C}(u, t)$ in which $a_n = c_0$, contradicting the valuation conclusion. Thus every such set is null. For a fixed ρ, if the one-dimensional set contained an interval, the same germ argument over $\mathbb{C}(t)$ would contradict the fixed-direction valuation conclusion. A semialgebraic subset of $\mathbb{R}$ without an interval is finite. The union over all finite words, all times and the three values c_0 is therefore null in the plane and at most countable on every fixed-direction slice.

This union includes every first extinction by applying the surviving word preceding it. Outside it, all states remain mixed and no step equals $2/3$. The proof does not assume tie boundaries are null or that a feasible word has interior.

9.2 A convergent step sequence would be captured

A curvature output exceeds $1/\sqrt{8}$, since $1 < l(r) < 4$. A growth decrease requires $\theta_n < 1/3$ and therefore follows curvature; its output exceeds $1/\sqrt{12}$. Hence

$$a_n \geq m := \min\{a_0, 1/\sqrt{12}\} > 0.$$

The universal ratio bound $\theta_n \leq q$ gives an upper bound as well: if $a_n \geq 1$, the curvature candidate is at most one; if $a_n < 1$, growth is at most q. Thus steps stay bounded.

Suppose $a_n \to A$. Then $A > 0$ and $\theta_n \to 1$. Infinitely many growth selections would give a subsequence of ratios tending to $\sqrt{5/3}$, a contradiction. Thus curvature is eventually strictly selected and

$$l(r_n) = (a_n^{-2} + a_{n+1}^{-2})/2 \to A^{-2}.$$

The monotonicity of l gives an extended limit for r_n. If $A = 1$, then $r_n \to 0$ while its multiplier $q_1(a_n) \to \infty$, impossible for an eventually positive sequence tending to zero. If $A = 1/2$, then $r_n \to \infty$ while its multiplier tends to zero, likewise impossible. Otherwise r_n has a finite positive limit, so $q_1(A) = 1$. The only positive solution is $A = 2/3$ and then $r_n \to 5/7$.

Only after this eventual strict-curvature conclusion may the memory coordinate be discarded. The autonomous pair map is

$$T(r, a) = \left(rq_1(a), (2l(r) - a^{-2})^{-1/2}\right).$$

At $z_* = (5/7, 2/3)$ its derivative is

$$J = \begin{pmatrix} 1 & 90/7 \\ -49/162 & -1 \end{pmatrix}, \qquad J^2 = -(26/9)I.$$

The positive-definite matrix

$$H = \begin{pmatrix} 49/162 & 1 \\ 1 & 90/7 \end{pmatrix}$$

satisfies $J^T H J = (26/9)H$. Taylor's theorem yields a neighborhood and $\lambda > 1$ such that

$$\|T(z) - z_*\|_H \geq \lambda \|z - z_*\|_H.$$

A tail converging to z_* must hit it exactly; otherwise repeated expansion contradicts staying in that neighborhood. Outside the exceptional set this is impossible because no finite step is $2/3$. Thus steps do not converge.

Finally the source convergence theorem applies to this globally smooth strongly convex quadratic for every positive initial step, giving $x_n \to 0$. To lift the null exceptional set to the original initialization use sign charts

$$(s, r, a) \longmapsto (s, \varepsilon s\sqrt{r}/2, a), \quad s \neq 0, \quad \varepsilon \in \{-1, 1\}.$$

They are local diffeomorphisms. Fubini's theorem and local Lipschitz preservation of null sets transfer the planar exceptional set to a three-dimensional null set; the coordinate axes are null too. This proves Theorem 6. There is no claim of one countable step set working simultaneously for all directions, or local finiteness of the exceptional union over all times.

10 Positioning and reproducibility

All results use the final NeurIPS 2024 Algorithm 2 with $\theta_0 = 1/3$ [1]. Its convergence theorem and inverse-square relation remain credited to that source. The earlier slow-rate background addresses the last-gradient question posed there; it is not an accusation of a false convergence theorem. The scalar finite-capture result, nonflat phase theorem, superlinear realization, regularity counterexamples, and mixed-direction noncapture concern distinct hypotheses and should not be conflated.

The constructions above are defined by convergent series and fixed interpolation formulas. Their proofs do not rely on finite numerical trajectories. Supplementary scripts in the accompanying relative directory `reproducibility/` evaluate finite prefixes for the two interpolation examples and the phase calculation; those finite checks are illustrations only. The planar theorem supplies no certified named rational initial step outside its exceptional set, no attractor or chaos classification, and no quantitative distribution of steps. The Dini/log-Dini distinction is a pair of sufficient conditions and counterexamples, not an optimal modulus classification.

References

[1] Y. Malitsky and K. Mishchenko, *Adaptive proximal gradient method for convex optimization*, Advances in Neural Information Processing Systems **37** (2024), Algorithm 2, Lemma 6 and Theorem 2. Final conference paper.

[2] J. W. Siegel and S. Wojtowytsch, *A qualitative difference between gradient flows of convex functions in finite- and infinite-dimensional Hilbert spaces*, arXiv:2310.17610v1 (2023). `https://arxiv.org/abs/2310.17610v1`. General slow-convergence background; not a theorem for the present adaptive recurrence.